

Memory Controllers for DDR Memories

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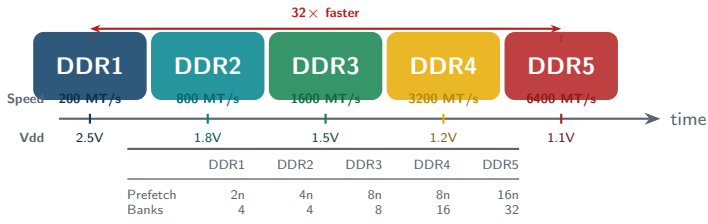
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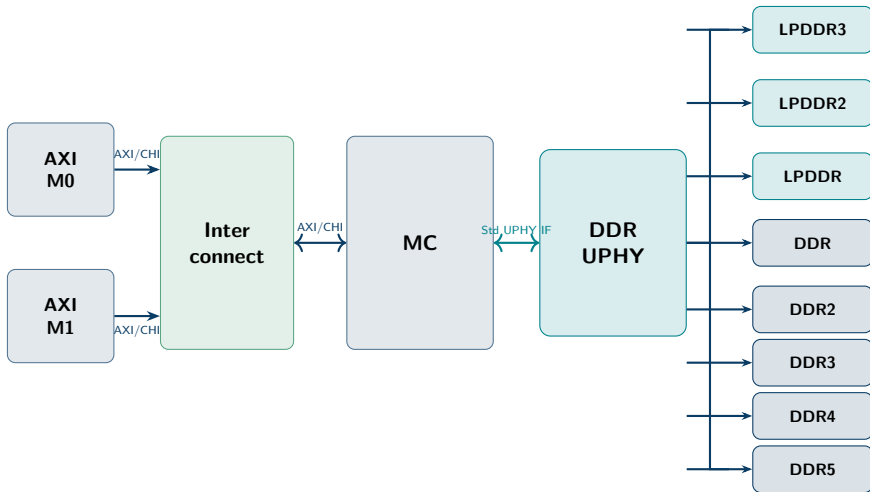
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Recap: Memory Controller for DDR Memories

- Acts as a bridge between the SoC and DRAM
- Enforces 50+ timing constraints (tWTR, tRC, tFAW...)
- Handles row-buffer management and request scheduling
- Manages power modes: PASR, TCSR, Deep Power-Down
- AXI out-of-order IDs map naturally to DDR bank parallelism

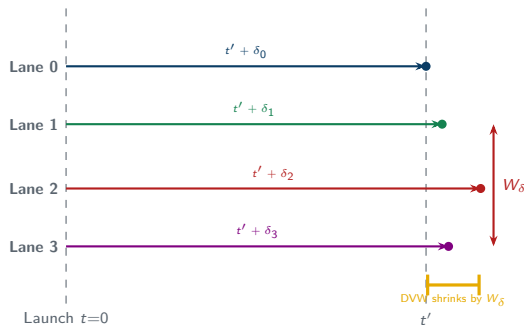


System Block Diagram



Inter-Lane Skew: Off-Die Signal Integrity

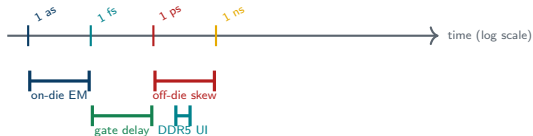
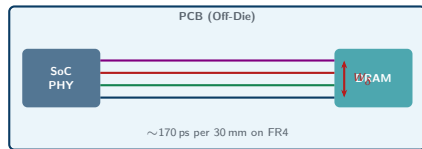
- DDR transfers data in parallel across multiple DQ byte lanes
- Each lane follows a different PCB path, picking up delay δ_i
- Skew window:
 $W_\delta = \max(\delta_i) - \min(\delta_i)$
- Receiver holds bus valid for the full W_δ before sampling
- DDR5 UI ≈ 156 ps; 20 ps skew eats >12% of it
- PHY training (Write Levelling, DQS Gating) corrects δ_i per lane



Off-Die vs On-Die Delay

Off-Die (Picoseconds)

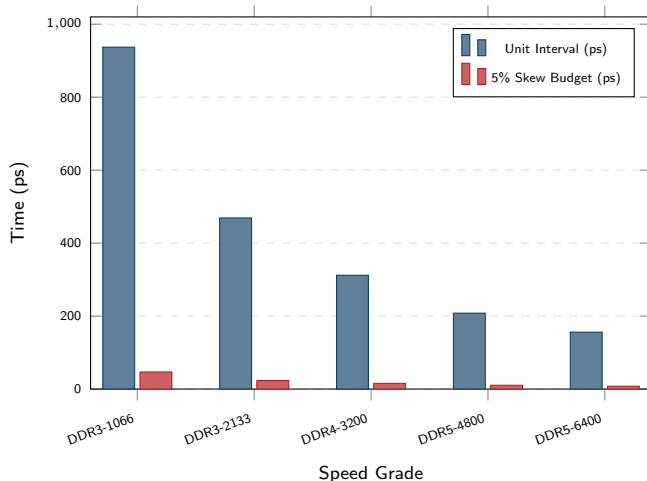
- Signals traverse PCB traces, vias, solder balls
- FR4 velocity $\approx 0.6c$; ~ 1 ps per $180\text{ }\mu\text{m}$
- 30 mm trace $\approx 170\text{ ps}$; W_δ typically $5\text{--}50\text{ ps}$



On-Die (Femto/Attoseconds)

- nm-scale on-chip metal interconnect
- $1\text{ }\mu\text{m}$ wire $\approx 3\text{--}10\text{ as}$ propagation
- Gate delay (3 nm): $1\text{--}5\text{ ps}$; bottleneck is logic depth

Why Picoseconds Matter at Modern DDR Speeds



Speed	UI	5% Budget
DDR3-1066	937 ps	46.9 ps
DDR3-2133	469 ps	23.4 ps
DDR4-3200	312 ps	15.6 ps
DDR5-4800	208 ps	10.4 ps
DDR5-6400	156 ps	7.8 ps

- At DDR5-6400 the skew budget is just **7.8 ps**
- Uncompensated off-die skew can fill this entirely
- PHY training trims δ_i to $\ll 5$ ps per lane

DFI: The Standard MC-PHY Interface

- **DFI** is the industry-standard interface between MC and DDR PHY
- Decouples MC from PHY implementation details
- Enables IP reuse across foundry nodes and vendors
- Covers command, data, training and power signals
- DFI 5.1: PHY-independent boot, expanded freq range, enhanced calibration feedback



Version	Key Addition	Support
1.0	Initial spec	DDR, DDR2
2.0	Training interface	DDR3, LPDDR
3.0	Low-power handshake	DDR4, LPDDR
4.0	DDR5 read/write paths	DDR5
5.0	PHY-independent boot	DDR5, LPDDR
5.1	Full back-compat + new ctrl IF	All gens

Three pillars of DFI 5.1

① **PHY-Independent Boot**

MC initialises DRAM without knowing PHY internals

② **Expanded Frequency Range**

Covers low-power LPDDR through DDR5-6400

③ **Enhanced PHY↔MC Interaction**

Real-time calibration feedback and margining

Backward compatibility

- Supports DDR through DDR5 and LPDDR through LPDDR5
- One DFI 5.1 MC paired with generation-specific PHY covers every standard

Why it matters for our MC

- Single RTL targets DDR3, DDR4, DDR5 and LPDDR4
- Swapping PHY = different memory standard, same controller
- Adopted by ARM, Synopsys, Cadence PHY IP

DFI 5.1 Architecture: Universal MC + Generation-Specific PHYs

